CITY BRAIN / FEATURE EXPLORATION

# What can we build?

## Cities: Skylines II - summary and possibilities report

City Brain can use the game as an interactive city environment: observe what happens, decide on a response, apply a change, and compare the result. The strongest starting point is a small traffic experiment with a dashboard and reliable measurements.

This report condenses the supplied 24-page technical study into feature tables. It retains all 36 scenario ideas and explains the work needed to explore them. The suggested order is a planning judgment, not a delivery estimate.

| Direction | Possible product experience | Suggested position |
|---|---|---|
| **City observatory** | See city health, traffic, services and selected buildings in a live dashboard. | **Start here.** Connect and validate the available observations. |
| **Traffic experiment lab** | Change one junction and compare congestion under the same starting conditions. | **First demonstration.** The study identifies a concrete bridge action. |
| **Daily-life simulator** | Explore commuting, school schedules, remote work and transport demand. | **Next.** Add schedule mods and keep demand settings stable. |
| **Emergency response lab** | Track fire response or explore access to services during flooding. | **After the foundation.** Incident tracking and control need adapters. |
| **Resilience research** | Explore outages, evacuation inequality, disease or detailed flood impacts. | **Longer-term.** Several ideas need external models. |

## How to read the feature tables

**N** = native game mechanism; **M** = community mod; **C** = custom code; **X** = external model. Effort: **Low** = mainly setup; **Medium** = adapter or integration work; **High** = a new model or deeper integration. These are relative effort bands from the source, not calendar estimates.

**Evidence boundary.** A native mechanism does not guarantee an automated trigger. The supplied study inspected documentation and source code but did not run the game/mod combination. This summary does not revalidate current compatibility. Treat the game as a simulation environment; real-world predictive accuracy needs separate evidence.

**Source:** Cities_Skylines_II_Twin_Layer_Report.pdf, revised 5 October 2026. Page references below refer to that original PDF. Overview: pp. 1-2, 17-20; original bibliography: pp. 21-24.

# 1. Core features to implement

These features form the reusable City Brain application. They make later scenarios observable, controllable and comparable.

Source: pp. 2, 5-12, 15-17. Effort and suggested order below are synthesis judgments.

| Feature | What it would do | Implementation and main limit | Order / effort |
|---|---|---|---|
| **Live city dashboard** | Show population, health, happiness, finance, service state and traffic diagnostics. | Use the community CS2MCP bridge; confirm loaded-city state and label game metrics correctly. | First<br>Medium |
| **City map and asset view** | Locate roads, buildings and services; inspect a small area. | Read positions and the road graph. Keep entity index + version and refresh after reload. | First<br>Medium |
| **Scenario runner** | Pause, advance a fixed interval and collect observations from a saved baseline. | Existing time/save routes help; saving is asynchronous. Verify completion and reset procedures. | First<br>Medium |
| **Junction controls** | Compare lights, no lights, stop control or default behavior at one junction. | The study identifies a bridge action. Read back the result; arbitrary green-time control is separate work. | First<br>Medium |
| **Results comparison** | Plot baseline versus intervention runs and retain settings and observations. | Custom logging and charts; record game/mod versions, frame time, run ID and missing samples. | First<br>Medium |
| **Incident timeline** | Display detection, dispatch, arrival and incident completion when observable. | Custom incident tracking and vehicle association. Notifications alone are not a full incident history. | Next<br>Medium-High |
| **Rules and response logic** | Apply a predefined response when a measured condition is met. | Custom rules and acknowledged commands. Each controlled mechanism needs one owner and a verified action. | Next<br>Medium-High |
| **GIS / neighborhood model** | Export assets and study access to essential services in a recognizable area. | Use the editor and Carto; validate coordinates, scale and heights. Automated OSM reconstruction is experimental. | Next<br>Medium-High |
| **Scenario-writing assistant** | Help describe experiments and explain measured outcomes in plain language. | Optional language-model layer grounded in recorded data; the source places it outside per-tick citizen control. | Later<br>Medium |

**Reliability work belongs in the foundation:** consistent snapshots, reload detection, bounded command queues, command IDs, read-back checks and stale-data handling. A successful HTTP request alone does not prove that a scenario is reliable. [Source: p. 11]

# 2. Movement and daily routines

Explore how travel demand and road controls change congestion, accessibility and emergency movement. Start with one intervention and keep the baseline stable.

Source: pp. 12, 15, 18. N = native; M = mod; C = custom; X = external. Effort includes the route shown, not a full product build.

| Possible feature | Experiment / explanation | What to measure | Route / effort |
|---|---|---|---|
| **01 Morning commute** | Tune work and school schedules to create a recurring demand peak. | Peak queues and sampled trip duration. | N + M<br>Low |
| **02 Staggered shifts** | Spread work start times to test whether demand becomes less concentrated. | Congestion and completed trips. | M<br>Low |
| **03 School arrival surge** | Vary school attendance or arrival windows around selected schools. | School access and surrounding road queues. | M + C<br>Medium |
| **04 Remote-work policy** | Change the share of people working remotely. | Travel demand and service side effects. | M<br>Low |
| **05 Event departure wave** | Schedule a venue departure surge with a custom event adapter. | Transit stop queues and road clearance. | M + C<br>Medium |
| **06 Junction redesign** | Compare junction controls or lane connections; verify each change. | Game traffic flow; add traversal and waiting measurements. | N + M<br>Medium |
| **07 Roadworks / closure** | Implement a reversible access restriction; demolition is not equivalent. | Detours and isolated destinations. | C<br>Medium |
| **08 Traffic collision** | Observe collisions; controlled spawning needs a verified event adapter. | Queues and disruption after an incident. | N + C<br>High |
| **09 Transit service cut** | Reduce vehicles on a selected public transport line. | Waiting and changes in travel mode. | N + C<br>Medium |
| **10 Freight bottleneck** | Constrain access to a cargo corridor. | Delivery delays and freight queues. | N + C<br>Medium |
| **11 Parking pressure** | Vary parking availability around destinations. | Destination access and circulation. | N + C<br>Medium |
| **12 Emergency corridor** | Test junction or access changes along an emergency route. | Response-time distribution for linked incidents. | M + C<br>High |

**Best first choice: 06.** The inspected bridge supports junction control modes. Exact signal phase timing and emergency-vehicle priority require further integration. Low-effort schedule experiments still need a compatible mod and a stable baseline.

# 3. Hazards and emergency scenarios

These possibilities range from observing existing game events to coupling specialist hazard models. Automated triggers and impact measurements must be verified separately.

Source: pp. 13-14, 19. N = native; M = mod; C = custom; X = external. A proxy is a simplified representation of a hazard.

| Possible feature | Experiment / explanation | What to measure | Route / effort |
|---|---|---|---|
| **13 Building fire** | Track a rehearsed fire; add a verified trigger later. No fire-start bridge route is documented. | Detection, dispatch, arrival and game damage. | N + C Med-High |
| **14 Forest-edge fire** | Use a validated ignition pathway near a district. | Response coverage and affected area. | N + C Med-High |
| **15 Hail disruption** | Use a verified event pathway to create disruption. | Road incidents and repair progress. | N + C Med-High |
| **16 Tornado evacuation** | Script a tested event and explore shelter access. | Evacuation completion and access. | N + C High |
| **17 River overflow** | Change a water source; map water depth onto roads and services. | Inundated assets and lost access. | M + C Med-High |
| **18 Coastal surge proxy** | Raise boundary water to represent coastal exposure. | Exposed roads and facilities. | M + C High |
| **19 Dam-release proxy** | Change upstream inflow; this does not model structural dam failure. | Downstream access loss. | M + C High |
| **20 Urban flash flood** | Couple external rainfall/runoff and drainage calculations to city impacts. | Flood footprint and access consequences. | X + C High |
| **21 Drought / low water** | Reduce available water sources and observe shortages. | Supply deficits and service effects. | M + C Medium |
| **22 Heatwave stress** | Vary temperature-related demand; model health exposure externally. | Demand stress and modeled exposure. | N + C + X High |
| **23 Snow / cold spell** | Test road maintenance and energy demand on a suitable map. | Road disruption and energy stress. | N + C Medium |
| **24 Earthquake / landslide** | Apply an externally defined damage footprint to the city. | Service disruption and access loss. | X + C High |

**Flooding needs an explicit impact rule.** Define when a road becomes impassable, account for bridges, and verify traversal actually stops. Visible water alone does not validate flood physics. SWMM is a runoff/drainage candidate; a 2D flood surface may require additional coupling.

# 4. Services and urban resilience

Explore how infrastructure failures affect residents and how the city recovers. The game provides service mechanisms; City Brain must define the event sequence and response policy.

Source: pp. 16-17, 20. N = native; M = mod; C = custom; X = external. Most controlled failures still need custom adapters.

| Possible feature | Experiment / explanation | What to measure | Route / effort |
|---|---|---|---|
| **25 Local blackout** | Disable a selected power supply path. | Buildings without power and restoration time. | N + C Medium |
| **26 Transformer bottleneck** | Constrain transfer capacity; compare alternate connections. | Supply constraints and network resilience. | N + C Medium |
| **27 Water outage** | Interrupt a water source or supply path. | Residents and essential services affected. | N + C Medium |
| **28 Sewage overload** | Reduce wastewater processing capacity. | Backlog and affected districts. | N + C Medium |
| **29 Pollution exposure** | Feed game pollution state into an external exposure model. | Modeled exposure, separately from game health. | N + C + X High |
| **30 Hospital surge** | Introduce a transparent synthetic wave of care demand. | Access and occupancy proxies. | N + C High |
| **31 Garbage backlog** | Reduce collection capacity, then restore it. | Accumulation and recovery time. | N + C Medium |
| **32 Supply interruption** | Constrain freight access to businesses. | Shortages and business operation. | N + C Medium |
| **33 Crime hotspot** | Define a synthetic service-demand concentration. | Service access under stated assumptions. | N + C Med-High |
| **34 Disease outbreak** | Run an external contact or compartment model. | Modeled spread, mobility and care demand. | X + C High |
| **35 Evacuation inequity** | Assign cohorts with different mobility or warning compliance. | Accessibility gaps and unmet demand. | X + C High |
| **36 Sensor / network outage** | Drop, delay or corrupt synthetic observations while retaining true game state. | Fallback decisions and sensitivity to missing data. | C Medium |

**Useful combined scenario:** flooded road → delayed ambulance → reduced hospital access → alternative route. Each transition needs an explicit model or adapter and a measurable outcome. The chain is a proposed experiment, not an automatically available game behavior.

# 5. A practical exploration path

Build one complete observation-to-action loop, then select an extension based on what the first experiment demonstrates.

Source: pp. 2, 7-17. The sequence below is a proposed implementation plan based on the supplied study.

| Stage | Deliverable | Evidence needed before expanding |
|---|---|---|
| **1. Observe** | Small saved city + bridge + dashboard for one busy junction. | Returned assets match the visible city; pause and reload are handled correctly. |
| **2. Intervene** | A junction-control action with a recorded request and confirmed result. | Read back the control state and inspect neighboring roads for displaced congestion. |
| **3. Compare** | Baseline A and intervention B from the same save and demand settings. | Repeat 3-5 pairs; show individual runs, averages and missing samples. This is not a statistical guarantee. |
| **4. Extend** | One schedule, incident, flood or sensor-failure scenario. | Its trigger, measurement and reset work reliably before adding another cause. |

## Choose tools according to the question

| Need | Candidate from the source | What still needs checking |
|---|---|---|
| Read / control the game | **CS2MCP** community bridge; optional MCP server. | Compatibility with the installed build; reliable observations and command confirmation. |
| Travel behavior | **Realistic Trips**; **RWH** for household/workplace capacity. | Introduce changes before the baseline; allow population and demand to stabilize. |
| Road control | **Traffic**; **Traffic Lights Enhancement**. | Lane edits, control ownership and mod compatibility; exact duration control is not established. |
| Water / geographic analysis | **Water Features** and **Carto**. | Automation adapters, coordinates, road passability and interpretation of depth data. |
| Specialist simulation | **SUMO / TraCI** for traffic; **SWMM** for runoff/drainage. | Synchronization, model ownership and validation; these are separate engineering projects. |

## Minimum evidence to keep

Record the baseline save hash, game build, mod versions, bridge revision, scenario settings, run ID, simulation frames, requested and confirmed actions, and missing samples. Keep game traffic diagnostics distinct from vehicles/hour; add instrumentation for actual trip or response times.

**Recommended first scope:** one neighborhood, one junction, a live dashboard, one verified control change, and a comparison report. Add fire-response tracking for an emergency focus, schedules for mobility, or sensor outages for decision robustness. Automated disaster creation, calibrated casualty estimates and exact replay are not established capabilities.

**Traceability:** bridge and integration details: original pp. 5-12; fire/flood: pp. 13-14; mods and geography: pp. 15-16; evaluation: p. 17; full scenario catalog: pp. 18-20; underlying references: pp. 21-24.